



Faster CNN-based vehicle detection and counting strategy for fixed camera scenes

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Abstract

Automatic detection and counting of vehicles in a video is a challenging task and has become a key application area of traffic monitoring and management. In this paper, an efficient real-time approach for the detection and counting of moving vehicles is presented based on YOLOv2 and features point motion analysis. The work is based on synchronous vehicle features detection and tracking to achieve accurate counting results. The proposed strategy works in two phases; the first one is vehicle detection and the second is the counting of moving vehicles. Different convolutional neural networks including pixel by pixel classification networks and regression networks are investigated to improve the detection and counting decisions. For initial object detection, we have utilized state-of-the-art faster deep learning object detection algorithm YOLOv2 before refining them using K-means clustering and KLT tracker. Then an efficient approach is introduced using temporal information of the detection and tracking feature points between the framesets to assign each vehicle label with their corresponding trajectories and truly counted it. Experimental results on twelve challenging videos have shown that the proposed scheme generally outperforms state-of-the-art strategies. Moreover, the proposed approach using YOLOv2 increases the average time performance for the twelve tested sequences by 93.4% and 98.9% from 1.24 frames per second achieved using Faster Region-based Convolutional Neural Network (FR-CNN) and 0.19 frames per second achieved using the background subtraction based CNN approach (BS-CNN), respectively to 18.7 frames per second.

Keywords Vehicle detection and counting · Convolution neural network · Fixed camera · Faster R-CNN · YOLOv2 · Feature point analysis

1 Introduction

Traffic information analysis is an important task in an intelligent transportation system (ITS) [5, 38] by offering efficient information for traffic control and management. Estimat-

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ing the number of vehicles running in video sequences is an important process in various applications used in selecting the best routes, manage the traffic lights, and help governments to make a decision about building new roads and plan the expansion of the traffic system. These numbers indicate the traffic status, including congestion level, lane occupancy, and road-traffic intensity [8]. Such kind of information can be utilized for automatic route planning, prevent road congestion, and early incident detection. In the traditional intelligent transportation systems, vehicle counting process is always implemented using special sensors. However, their higher installation cost and the simple format represent some limitations for those sensors.

With the growth of digital video processing, counting system based on image processing and computer vision techniques, offers an attractive alternative method for its large ability in detecting vehicle type, velocity, density, and road traffic accident [3]. The machine vision vehicle counting method is an integrated procedure comprised of detection, tracking, and trajectory processing. Vehicle detection is considered the first step in obtaining traffic flow characteristics [34]. Its purpose to obtain the location and classification of the object from an image [13]. Its primary task is to acquire the features of the object. Traditional detection approaches based on low ranks decomposition [31, 37], gaussian mixture modeling [40], Morphological operations [12], and principal component analysis [24] can be used for the detection. It exhibits better performance if the object has some deformation or scale change. However, these methods cannot adapt to large rotations, has poor stability, and is slow to calculate. In recent years, Deep Learning strategies, more specifically convolutional Neural Networks (CNNs) [10, 11, 17, 20, 25], have shown significant advances over the traditional approaches in many computer vision tasks, including object detection and classification.

Detection and tracking of vehicles are normally done as two separate procedures. Vehicle detection in images relies on features of spatial appearance, while vehicle tracking relies on features of spatial appearance as well as temporal motion [6, 9, 13]. Vehicle detection techniques focused on visual and appearance characteristics of vehicles' spatial domain have begun to saturate. Different methods have been introduced in the literature to address the detection and counting problems of vehicles in images and videos. However, due to the changes in shape, scale, object view, shadows, lighting conditions, and partial occlusion, it is still an open issue. Although substantial advances have been made separately for detection or tracking, most of them still require a high computation complexity with low accuracy.

Not much effort has been made to use tracking information to increase object detection and counting accuracy and minimize storage and training costs. In addition, the object detector usually suffers from missed detection and false positives that deteriorate the process of counting. Due to such issues, this work proposes a successful collaborative strategy between detection and tracking information to strengthen both detection and counting processes. Detection and tracking of vehicles simultaneously are considered an efficient technique for achieving precise detection and counting results. Firstly, we use the transfer learning for the YOLO-v2 vehicle detector as an initial step to discriminate between vehicles and other foreground and background regions. The second point of our contribution is to refine the detection result to achieve a perfect vehicle detection result, the optical flow tracking information of the detected feature points is used to achieve an accurate vehicle detection. The third part of our contribution is to assign each vehicle with its corresponding trajectory based on the temporal information of the refinement detection and tracking feature points between the framesets to achieve better detection and counting results.

Moreover, to enhance both the time performance and the overall detection and counting accuracy, different CNN-based detections, including BS-based CNN, FR-CNN, and YOLO

v2, were tested in the detection part with the tracking information help. These methods' overall performance is analyzed in terms of detection, counting, and time performance accuracy. Also, to highlight our contribution about the importance of using both the detection and tracking information in the detection decision, we evaluated the detection accuracy in two cases

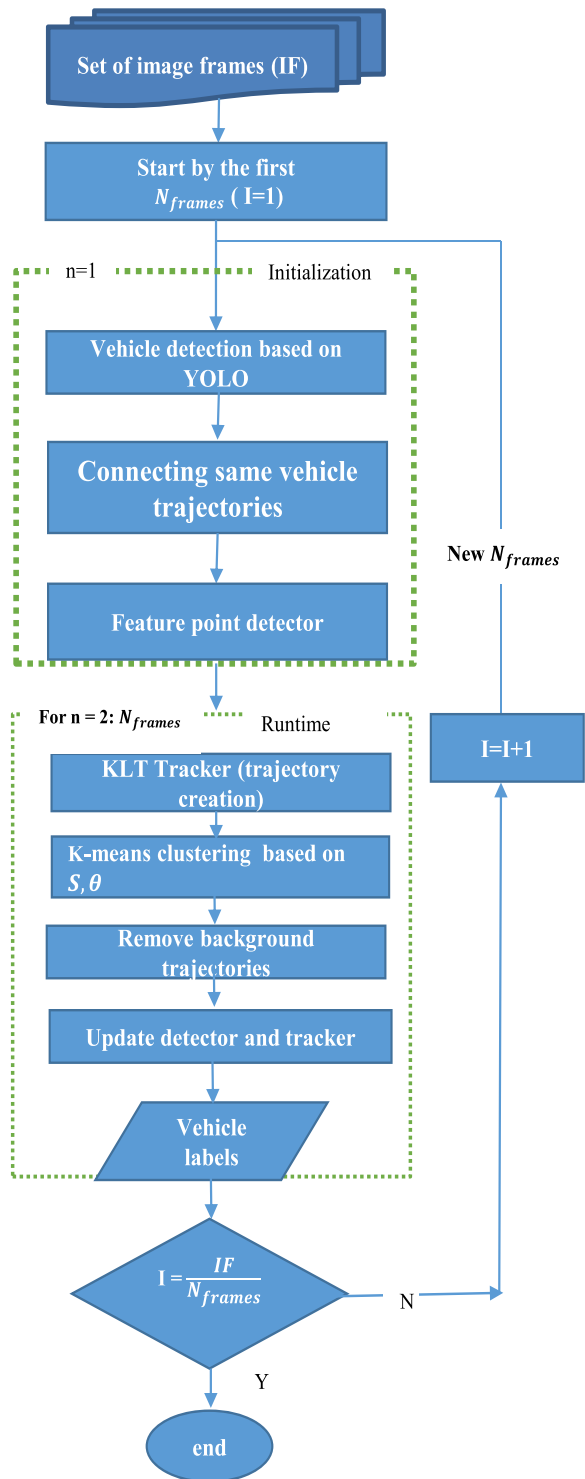
- Detection decision is based only on the YOLO-v2 and the detection implemented on every frame.
- Detection decision is based on the YOLO-v2 and tracking information, and the detection is implemented once on the first frame, every fixed number of frames.

This work used the detection information of YOLOv2 based CNN, and tracking information using optical flow. The first step contains vehicle detection and features extractor, while the second step to analyze these features and vehicle refinements. First, the power of the convolution neural network is exploited in the vehicle detection process before the vehicle refining and clustering process in the second step using the optical flow and k-means clustering. The CNN is used in the first frame, with the refining analysis considering the remaining frames in the frameset. Thus, a robust discrimination process between the foreground vehicles and noisy background regions is utilized. Thirdly, an effective counting strategy is offered to assign each vehicle with its corresponding trajectory based on the collected detection and tracking information. Moreover, we consider the performance evaluation of three categories of CNN architectures in the context of vehicle detection, in terms of accuracy and processing time. We conducted experiments on challenging datasets, and the proposed method showed the best performance in terms of precision and recall. The proposed framework contains three functional steps in each frameset, as shown in Fig. 1. In summary, this study provides the following four main contributions:

- Rapid and reliable vehicle detection and counting strategy is presented. The developed strategy exploits the regression based CNN benefits and the optical flow information to obtain a faster and more reliable result.
- Comprehensive analysis of the performance of three categories of CNN architectures used in the detection part in cooperate with tracking information in the context of vehicle detection and counting is presented, in terms of accuracy and processing speed.
- The performance evaluation of two detection based categories is presented, the first one when the detection decision is based only on the YOLO-v2 and the second when the detection decision is based on both the YOLO-v2 and tracking information.
- Extensive experiments on four benchmark datasets demonstrate the effectiveness of the proposed strategy and its ability to achieve faster detection and counting results with efficient accuracy.

The main steps of proposed vehicle detection and counting scheme are described in the following subsections. The rest of this work is organized as follows. Section 2 summarizes the related work. Section 3 presents the proposed strategy. Section 4 reports the experimental results and discussion. Section 5 demonstrates the conclusion.

Fig. 1 Architecture of the proposed system (n is the frame number in video)



2 Related work

In recent years, several deep learning networks have been utilized in vehicle detection and counting. A classification framework based on deep convolutional neural network (DCNN) was previously reported. Its accuracy has been greatly improved compared with traditional classification algorithms, which lays a foundation for deep learning-based object detection research. However, only detection information has some false positive and negative results. Ross Girshick et al. used a selective search algorithm named regions with CNN (RCNN) based on region proposal strategy that searches for possible object regions [11]. According to the slow detection speed of RCNN, Ross Girshick et al. proposed Fast RCNN [10], by adding bounding box regression and multi-task loss function. In [27], the authors added a new region proposal network based on the fast RCNN algorithm and proposed the Faster RCNN. The accuracy of the Faster RCNN has been greatly improved and is rated the best in all current detection algorithms, but the speed is one of its drawbacks. To solve this problem, Liu proposed an end-to-end detection algorithm, a single-shot multi-box detector (SSD) [20], which obtains proposal regions by uniform extraction and greatly enhances the detection speed.

In the most recent year, Redmon et al. proposed YOLOv2 [25], and YOLOv3 [26] by using multi-scale prediction and improving the basic classification network, with fast detection speed. In [1], authors present YOLOv4 that utilizes an additional bounding box regressor based on the Intersection over Union (IoU) and a cross-stage partial connections in their backbone architecture, and used mosaic and cut-mix data augmentation. However, all of the mentioned algorithms depends only on the detection decision each frame, so the accuracy is still needs improvement with fast processing speed approach.

Authors in [4], presented a new vehicle detection and counting strategy by employing a real-time background model, filling holes and denoising optimization, and motion-based information analysis. This strategy can handle vehicle shadows and sudden illumination changes. Moreover, the counting accuracy has been enhanced by using the ROI concept called Normative-Lane and Non-Normative-Lane. However, their use of traditional background modeling and the morphological operation will affect the results of the algorithm and deteriorate the detection and counting performance. Song et al. proposed a vehicle detection and counting system by employing YOLOv3 for detecting the vehicles. Then, the ORB approach [28] was adopted for the vehicle trajectories and counting [33]. This algorithm achieved a satisfying accuracy, but it still needs more time improvement because it depends on the feature point purification to exclude the false noise points using the RANSAC algorithm by estimated the homography matrix. Authors in [30], present a combined detection and tracking system that uses YOLOv3 for object detection and the Deep SORT for object tracking. They achieved better detection and counting accuracy, but their algorithm still requires more time.

Recently, the Convolutional neural network has achieved high performance for object detection, while dealing with CNN efficiently for real-time vehicle counting is still a challenging problem. In [14], authors combined the detection based CNN decision and tracking information to solve this problem. The algorithm firstly detects the vehicles using a CNN-based classifier with connected component labeling, then vehicle feature motion is analyzed to remove the noise and cluster the vehicles. Vehicle refinement has been done using a new combined strategy between the K-means clustering and optical flow tracking information. Finally, a way to assign the detected vehicles with its corresponding cluster is introduced, to ensure a non-repeated counting process, by considering the intersection area between

the detected and tracked point information. However, this algorithm achieved a promising result using the previously mentioned technique, but the processing time still needs to be reduced since the strategy employing a pixel classification CNN-based strategy for the detection. Various background subtraction based CNN strategies are introduced to overcome the quite slow for the patch-wise based methods by considering the full image as an input [21, 39]. However, background subtraction strategies cannot classify and differentiate the object class including vehicles, bicycles, or persons. So in this work, we evaluate different object detection and classification strategies like Faster R CNN and YOLOv2, with the tracking information help because we focus on vehicles in this work.

As concluded from the evaluation process, we present an efficient and fastest strategy for vehicle detection and counting. The proposed algorithm detects moving vehicles based on YOLOv2 classifier. Then, the vehicle's robust features are refined and clustered by motion feature points analysis using a combined technique between KLT tracker and K-means clustering. Finally, an efficient strategy is presented using the detected and tracked points information to assign each vehicle label with its corresponding one in the vehicle's trajectories and truly counted it.

3 Proposed methodology

The proposed method divided into three steps. First, Vehicles are classified and detected every N-frames using YOLOv2. Then, their features are extracted and analyzed by tracking and clustering the corner points through the N-frames using K-means clustering and KLT tracker. Finally, an efficient strategy is presented to assign vehicle trajectories to each detected bounding box, so that each vehicle trajectories will take a unique label. The architecture of the proposed approach is shown in Fig. 1. The main steps of proposed vehicle detection and counting scheme are described in the following subsections.

3.1 Vehicle detection

Recently, different deep learning-based strategies are used to enhance the detection result such as Single Shot MultiBox Detector (SSD) [20], region proposal approaches [11, 27], and You Only Look Once (YOLOv2) [25]. However, authors in [14] achieved a promising result using background subtraction based CNN for the detection with the help of tracking information, but the processing time still needs to be reduced since they used a pixel classification CNN-based strategy for the detection.

Recently, It is well known that YOLOv2 is one of the fastest object detection methods based on CNN compared to the state of the arts [23], so in this work, we investigate YOLOv2 in the detection phase combined with the tracking information to achieve faster detection and counting strategy. YOLOv2 is trained on more than a million images from the ImageNet database that consists of 1.2 million images classified into 1000 classes [29]. In our work, the YOLOv2 layers are used up to the last fully connected layer that changed from 1000 classes to two classes as we focus only on vehicles detection.

In this work, we exploit the Yolov2 architecture to achieve faster detection result employing the transfer learning. In addition using the optical flow information to improve the detection performance with the same counting strategy explained in [14] to achieve perfect counting decision. Transfer learning is implemented on the final layers by replacing the softmax 1000 classes by softmax two classes. In transfer learning, pre-trained convolution

neural network models are used which trained using big datasets. pre-trained model layers are used up to the last fully connected layer that is trained in our work using the dataset associated with vehicles. More details of the transfer learning method appear in the work of [22]. Since we utilize the transfer learning in this work, we used the Resnet-50 [16] as a pertained neural network model and considered a backbone for the YOLOv2. A block diagram illustrating the ResNet50 transfer learning architecture is shown in Fig. 2. ResNet50 is a CNN that trained on more than a million images from the ImageNet database. This database contains 1.2 million images classified into 1 thousand classes. Three training experiments were conducted in this work using different images collection, to investigate the performance of our proposed method using YOLOv2.

Fine tuning YOLO-v2 achieved a satisfactory recall accuracy. Nonetheless, it detects some of the false-positive objects that yield low precision accuracy. These false-positive results will be eliminated by employing K-means clustering and their tracking optical flow information, as described in the following subsections. The false-positive results obtained from the background regions have different motion characteristics compared to the foreground vehicles. Hence we exploit their feature points motion information to discard them before the final decision. There are different clustering strategies that can be used to group the feature points as mentioned in [7, 41, 42]. However, in this work, K-means clustering is sufficient to achieve a good accuracy with a low computation complexity.

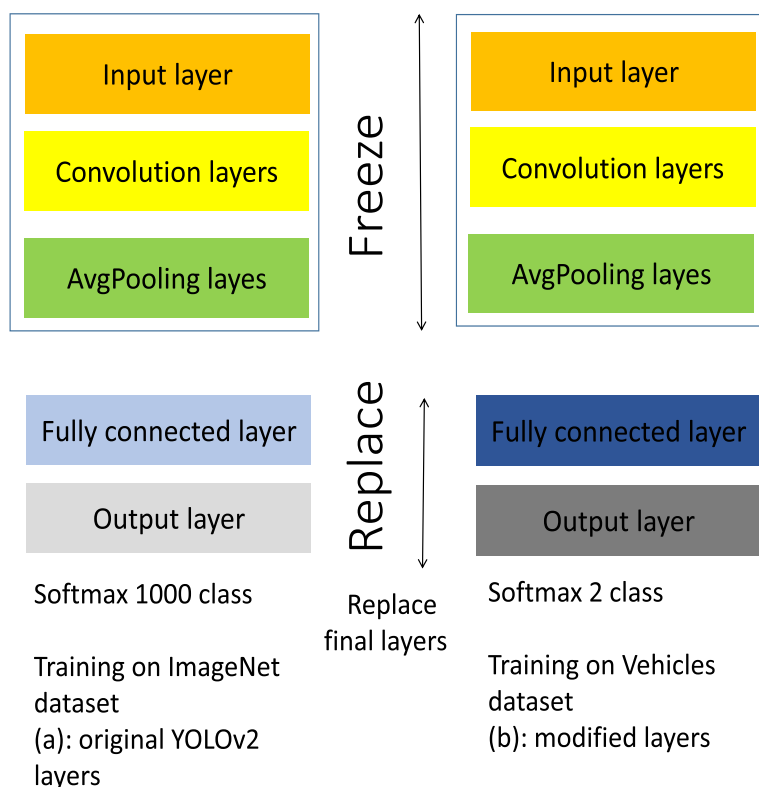


Fig. 2 Diagram represents the ResNet-50 transfer learning architecture used for vehicle detection

3.2 Vehicle features refinement and clustering

This step is used to extract the vehicles and eliminate the background regions, in addition to vehicles clustering process. Optical flow-based tracking leads to a faster processing speed and feature matching accuracy. So, optical flow Kanade-Lucas algorithm [2] is used for tracking the feature points from frame f to $f + 1$.

The result of the optical flow in the first frame pairs is a set of vectors C with elements $C_i = (S_i, \theta_i)$, where S and θ are given by

$$S_i = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2} \quad (1)$$

$$\theta_i = \arctan\left(\frac{Y_2 - Y_1}{X_2 - X_1}\right) \quad (2)$$

where X_1 and Y_1 represent the X and Y coordinates in the previous frame, while X_2 and Y_2 represent the X and Y coordinates in the next frame. Each element in C corresponds to an interest feature point P_i tracked from frame f to $f + 1$, where S and θ are two vectors containing the displacement magnitudes and angles respectively for each corner point.

The noisy detections tend to result in short-lived trackers. So in this work, the foreground detection is considered a vehicle object only if it tracked in several consecutive frames (9 consecutive frames) since the noisy detections may be tracked for short-lived period. Then k-means clustering is used to cluster the remaining foreground vehicles. More details can be found in [14].

3.3 Vehicle counting

After extracting the most robust features and grouping them into separate clusters for each detected vehicle. Each of these vehicle features is assigned a unique ID and tracked until disappearing through the video. The assigning process is based on the intersection area of the old rectangular tracked bounding box and the new detected bounding box. If the intersection area is greater than a predetermined percentage α , this vehicle has the same identity of the old matched vehicle. While there is no intersection area or having an intersection area less than α , the vehicle gets a new label. There are four counting possibilities for each N-frames:

- A new vehicle detected for the first time, so all features inside its bounding box haven't assigned to a label. In this case, a new label is given to these features and the counter is increased by one.
- A vehicle bounding box is detected at the first frame and it was previously detected in the previous frameset (N-frames), thus some or all of its features have a label. In this case, the unlabeled features (if existed) take the same label.
- A vehicle bounding box is not detected in the first frame but it is detected before in the previous frameset. In this case, its features were given a label before, so the same label is continuously assigned to them.
- A vehicle is never detected through all the video framesets, this vehicle is represented as a missed counted vehicle.

Table 1 Training experiment details

Experiment number	Source of training images	Number of training images
Experiment I	Matlab vehicle images	259
Experiment II	highway baseline scene	127
Experiment III	Matlab images, intermittenpan, street corner at night, and tramstation scenes	less than 10% of the frames of each scene

4 Experimental results

To investigate YOLOv2 in the detection and counting strategy under different environment, three training experiments were employed as shown in Table 1.

All experiments, including twelve videos with various challenges are used to validate the contribution of the proposed method. The Videos consist of two sequences form GRAM dataset [15], M-30 and M-30 HD, four sequences form CDnet 2014 [35], HighwayII video from ATON Testbed, and five sequences from the public urban traffic data set UA-DETRAC [36]. We test the proposed approach on nighttime, daytime, intermittent vehicle motion, and crowd scenes as mentioned in Table 2. In all experiments, the fixed number of frames in each frameset is equal to ten frames, $N = 10$ for achieving better tracking and counting result using KLT [32].

We examined the algorithm with different values of α , where we found that if α is too high, the same vehicle may be classified into a new vehicle. The value of $\alpha = 25\%$ yields to the best accuracy in our experiment.

Quantitative evaluation of the detection and counting will be discussed and compared with the state of art approach [14]. The detection accuracy is evaluated using quantitative performance metrics that have been used as a standard evaluation [18], known as Precision

Table 2 Challenge environments information of the videos used in the performance evaluation

Dataset	Sequence	Challenging description
GRAM Dataset	M-30	Sunny day, and low resolution camera.
	M-30-HD	Cloudy day, and high resolution camera.
CDnet2014	Highway	Sunny day, shadows, and waving trees.
	Intermittenpan	Sunny day, and waving trees.
	Street corner at night	Light changes, and night scene.
	Tramstation	Night scene, and light changes.
ATON Testbed	Highway II	Crowded scene
UA-DETRAC	MVI-40852	Cloudy scene
	MVI-39401	Sunny day
	MVI-40772, MVI-40775, and MVI-40793	Night scenes



Fig. 3 Sample Matlab training images used in the fine-tuning

and Recall. The precision is calculated as the percentage of correctly detection vehicle pixels [true positive (TP)] over the total number of detecting object pixels including TPs and false positive (FP).

$$Precision = \frac{|TP|}{|TP| + |FP|} \quad (3)$$

Recall refers to the ratio of accurately detected vehicle pixels to the number of actual vehicle pixels that include the number of false negative pixels (FN).

$$Recall = \frac{|TP|}{|TP| + |FN|} \quad (4)$$

The counting precision can be defined as

$$Precision (\%) = 100 - Error (\%) \quad (5)$$

Where

$$Error (\%) = \frac{|Estimated - TrueNo. |}{TrueNo.} \times 100 \quad (6)$$

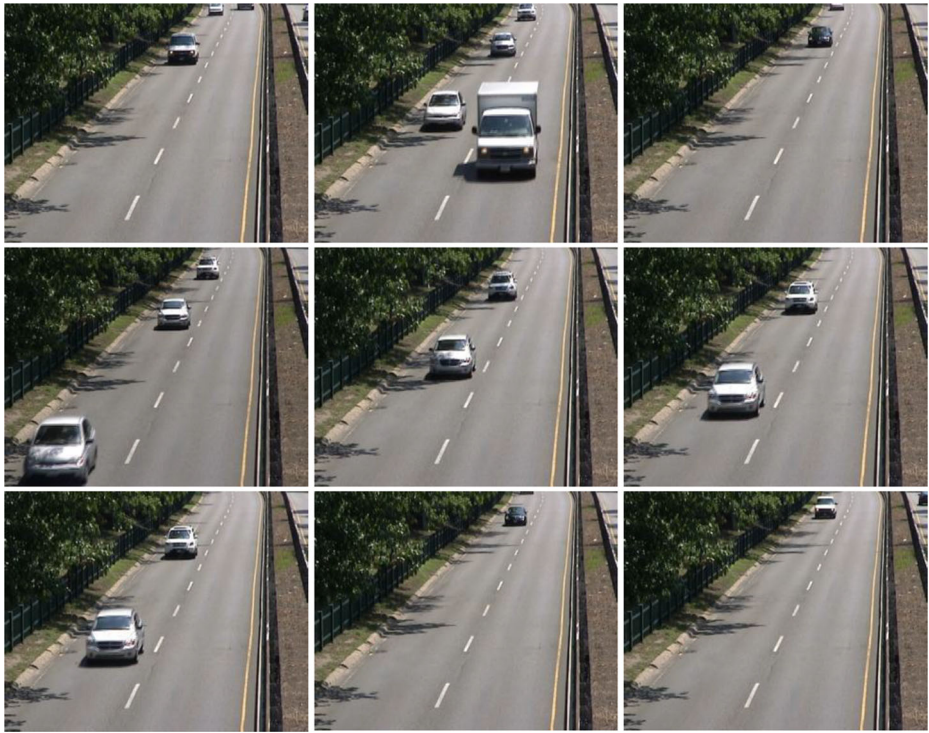


Fig. 4 Sample Highway training images used in the fine-tuning from CD-net2014 dataset

Moreover, the processing time of detection and counting is calculated to evaluate the computational time complexity of the proposed approach, represented by Frame Per Second (FPS). Time improvement is estimated according to the following equation

$$\text{Time Improvement (\%)} = \frac{|\text{Estimated FPS} - \text{Old FPS}|}{\text{Estimated FPS}} \times 100 \quad (7)$$

Where *Estimated FPS* is the number of processing frames using the proposed strategy, and *Old FPS* is the number of processing frames using the other compared strategy.

In experiment I, the YOLOv2 network is re-trained on Matlab collection of vehicle images, 295 images. Samples of these images are shown in Fig. 3. In experiment II, for the sake of comparison with background subtraction based proposed method [14], 127 frames of highway baseline scene in ChangeDetection.net dataset (CDnet 2014) are used to re-train the network. Samples of these images are shown in Fig. 4.

In experiment III, other datasets were used for enhancing the accuracy, frames collection from ChangeDetection.net dataset (CDnet 2014) including, Intermittenpan, street corner at night, and tramstation videos are used for the retrain the network. Samples of these images are shown in Fig. 5. It is important to note that the image frames used for training are not used also for testing. In experiment III, we used only less than 10% of the frames of



Fig. 5 Sample training images collected from matlab and CD-net2014 dataset sequences, used in the fine-tuning

different sequences including Tram station, Street corner at night, and Intermittenpan with the matlab images. While the remaining frames about 90% of these sequences are used in the testing. Moreover, In experiment III, we used the M-30 and M-30-HD sequences from GRAM dataset, highwayII sequence from ATON dataset, and five sequences (MVI-39401, MVI-40852, MVI-40772, MVI-40775, and MVI-40793) from UA-DETRAC dataset for testing, and these sequences are not included in the training images. The experimental results were implemented employing MATLAB on Intel i7-4810MQ CPU 2.80 GHz, 16 GB RAM, and Quadro K1100M GPU with 2 GB of video RAM. In addition, an experiment was conducted to test the performance evaluation using either CNN-based YOLOv2 for the detection or both of Yolov2 detection combined with optical flow tracking. We use the four videos of Changedetection.net 2014 (CD2014) dataset that mentioned in [14], with the same performance metrics.

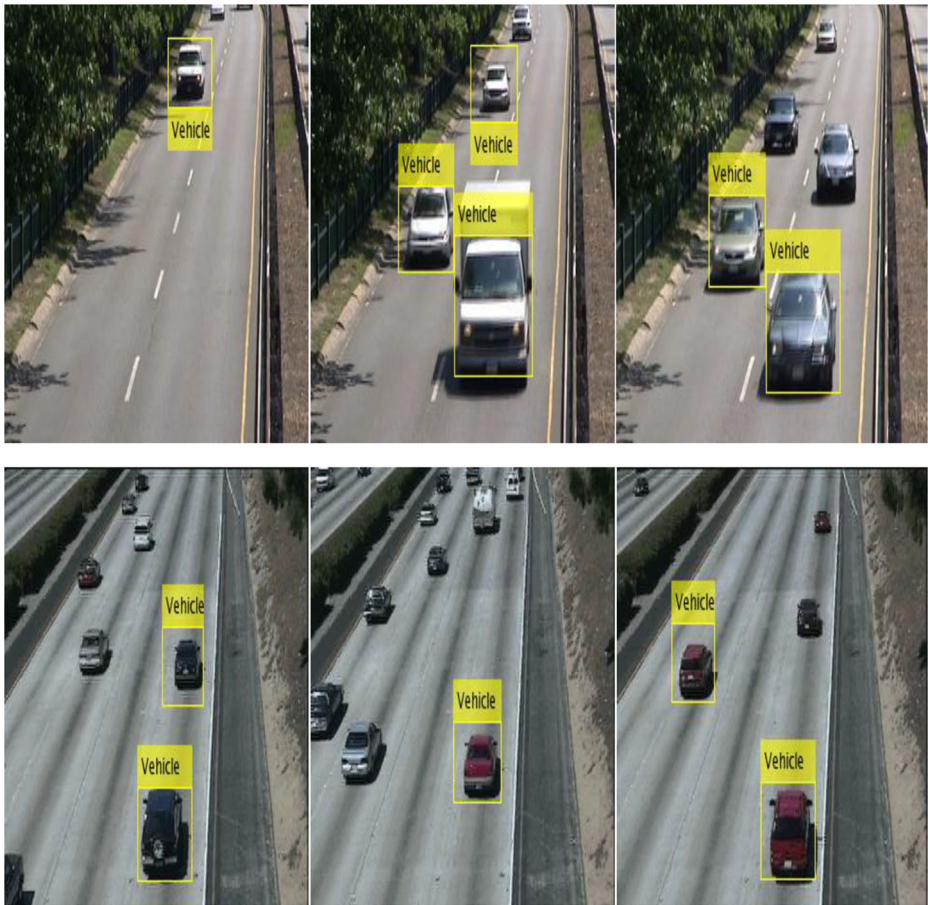


Fig. 6 Sample results show missing vehicles using experiment I

There are several pre-trained models available, but direct application to another dataset may not be feasible. The result may contain a lot of missed detections and some incorrect detections have been classified. Nonetheless, these pre-trained weights are still useful and can be used with a transfer learning approach to weight initialization and fine tuning of the network on a new dataset using a transfer learning approach.

In experiment I, the images used for training have an approximate visual feeling with the highway sequence from CDnet2014 dataset and highwayII sequence from ATON dataset, so YOLOv2, in this case can detect the vehicles from these two sequences but still the accuracy is not sufficient because some missed detections occur as shown in Fig. 6.

We evaluate the proposed methods in the two cases, the first one using background subtraction based CNN in the detection part, and the second case, using YOLOv2 based detection. As clear from Table 3, the proposed method using YOLOv2 and tracking information achieved the fastest processing speed with the lower counting precision compared to

Table 3 Vehicle counting results comparison on CDnet2014 sequence and ATON for experiment I

Compared Algorithm			BS-based Method [14]	YOLOv2-based Proposed Method
CDnet 2014 Dataset	Highway	Miss detection	0	2
		Precision	100	91.7
		Time	0.18	19.35
		Performance (FPS)		
ATON Testbed	Highway II	Miss detection	1	33
		Precision	97.9	31.25
		Time	0.18	29.7
		Performance (FPS)		

the method based on BS-CNN in the detection phase. It is asserted that the YOLOv2 based detection increases the time performance by 99.1% and 99.4% for Highway and Highway II respectively, according to (7).

In experiment II, to improve the performance, the training images are selected from one evaluation testing scenes, 127 frames of Highway sequence from CDnet 2014 dataset. Images used for training have an approximate visual feeling with the M-30 and M-30-HD sequences from GRAM dataset and highwayII sequence from ATON dataset, so YOLOv2 in this case can detect the vehicles from these three sequences but still the accuracy is not sufficient for GRAM dataset sequences and ATON dataset sequence, because some missed detections occur as shown in Fig. 7.

We compare the proposed method in the two cases, the first one using background subtraction based CNN (BS-CNN) in the detection part, and the second case, using YOLOv2 based detection. As clear from Table 4, the method using YOLOv2 and tracking information achieved the fastest processing speed with the lower counting precision compared to background subtraction based CNN in the detection phase in CDnet2014 dataset, and ATON dataset sequences. It is asserted that the YOLOv2 based detection increases the time performance by 98.9 %, 99.3 %, 98.9 %, and 99 % for M-30, M-30-HD, Highway II, and Highway respectively. However, using the background subtraction based CNN detection case, the algorithm has the best counting precision accuracy while the same accuracy achieved when the training images from the testing scene as mentioned in Table 4 for Highway sequence.

Tables 3 and 4 show that the YOLOv2 has lower accuracy when trained on images less visually to the testing images since vehicles from two scenes using experiment one can be detected while vehicles from four scenes using experiment II can be detected. Hence in experiment III, we increased the images used for training from different scenes to improve the accuracy.

In experiment III, we used only less than 10% of the frames of different sequences including Tram station, Street corner at night, and Highway with the images of experiment I to train YOLOv2 with different visual images from the testing videos. In this case, the images used for training have an approximate visual feeling with the testing dataset, so YOLOv2

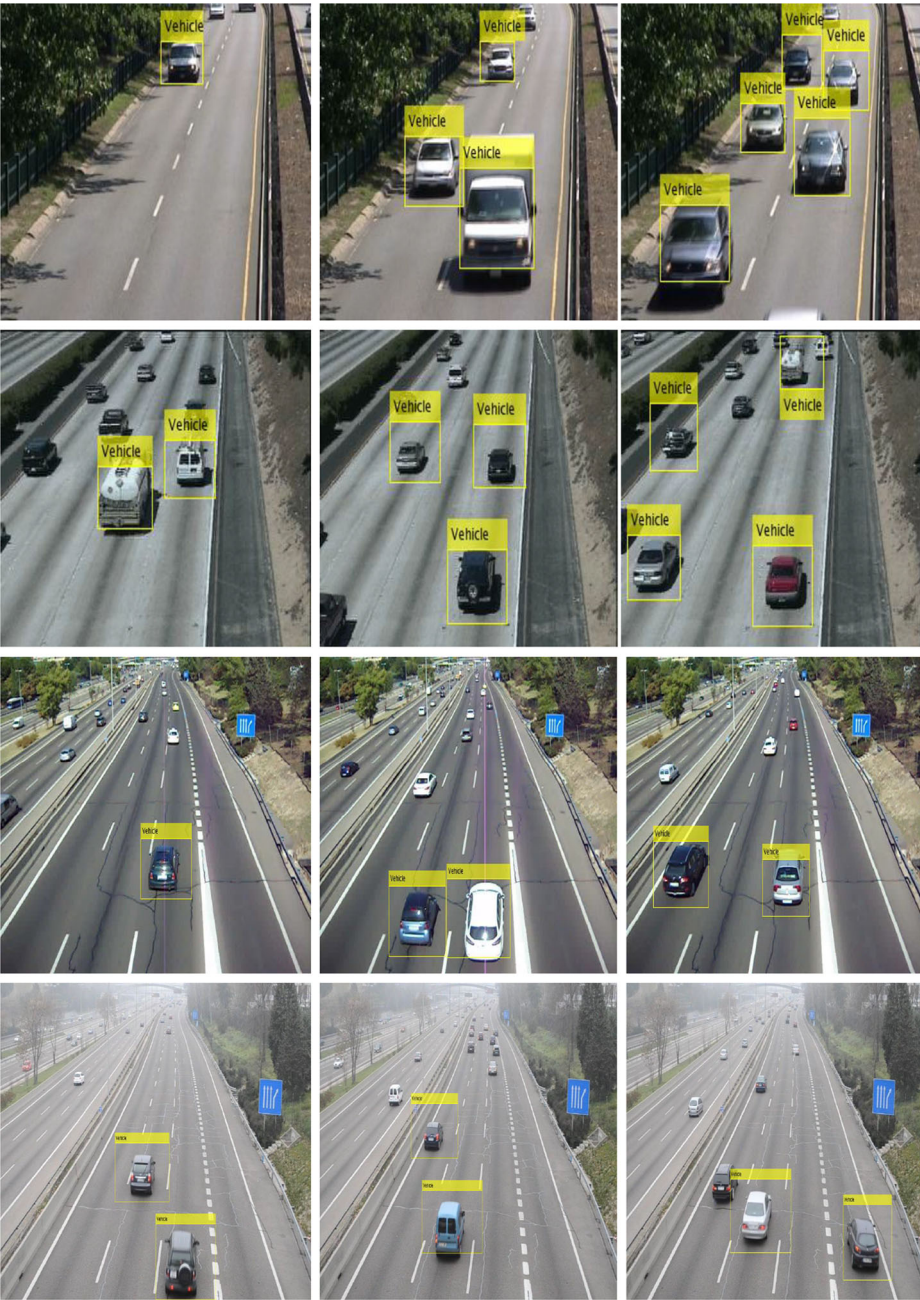


Fig. 7 Sample results show missing vehicles using experiment II

Table 4 Vehicle counting results comparison on CDnet2014, GRAM Dataset, and ATON for experiment II.

Compared Algorithm			BS-based Method [14]	YOLOv2-based Proposed Method
GRAM Dataset	M-30	Miss detection	0	35
		Precision	100	54.6
		Time	0.225	19.8
		Performance (FPS)		
	M-30-HD	Miss detection	0	23
		Precision	100	45.3
		Time	0.18	26.1
		Performance (FPS)		
	ATON Testbed Highway II	Miss detection	1	9
		Precision	97.9	81.3
		Time	0.18	17.55
		Performance (FPS)		
CDnet 2014	Highway	Miss detection	0	0
		Precision	100	100
		Time	0.18	18.9
		Performance (FPS)		

based proposed method results in this case improved comparing to the previous experiments as shown in Figs. 8 and 9.

We compare the proposed method in three cases, the first one using background subtraction based CNN (BS-CNN) in the detection part, the second case, using faster R-CNN based detection, and in the third case using YOLOv2 based detection. In addition, we investigate the algorithm when the detection is based only on the YOLOv2 algorithm compared to the detection decision based on Yolov2 with the tracking information as shown in Table 9. As clear from Table 5, the method using YOLOv2 and tracking information achieved the fastest processing speed compared to BS-CNN and faster R-CNN used in the detection phase, for CDnet2014 dataset sequences. The evaluation experiment shows that both of faster R-CNN-based proposed method and YOLOv2-based proposed method have a comparable precision and recall results.

It is asserted that the YOLOv2 based proposed method increases the average time performance by 94.9% from 0.81 frames per second of BS-CNN case to 15.75 frames per second. However, using the BS-CNN based proposed method, the algorithm average recall accuracy outperform by 13.2%, while the average precision accuracy reduced by 3.75%. The time improvement resulted from YOLOv2 based proposed method is due to the frame testing processing without batch processing. While the accuracy improvement using BS-CNN based proposed method is due to pixel classification for all frame pixels with more time.



Fig. 8 Sample detection results using experiment III. Top row: crowded and Cloudy. Second and fourth row: different camera view, sunny day, and waving trees. Third and Bottom row: Night scenes with changed light



Fig. 9 Sample counting results using experiment III. Top row: crowded and Cloudy. Second and fourth row: different camera view, sunny day, and waving trees. Third and Bottom row: Night scenes with changed light

Table 5 Vehicle detection results comparison on CDnet2014 sequences

Method		Yang et al. [37]	BS-based Method [14]	FR-CNN-based Proposed Method	YOLOv2-based Proposed Method
Sequence videos					
	Highway				
	Precision %	91.3	98.2	100	100
	Recall %	92.1	99.2	95	92
	Time Performance (FPS)	12	0.18	1.31	18.9
Intermittenpan	Precision %	90.2	98.7	100	100
	Recall %	98.8	97.5	83	81
	Time Performance (FPS)	12	0.27	0.49	13.95
Streetcorneratnight	Precision %	89.1	95.8	100	100
	Recall %	97.2	97.1	78	76
	Time Performance (FPS)	12	0.225	0.63	14.85
Tramstation	Precision %	87.1	92.3	100	100
	Recall %	97.1	97.1	93	89
	Time Performance (FPS)	N/A	0.18	0.72	15.3
Average accuracy	Precision %	89.4	96.25	100	100
	Recall %	96.3	97.7	87.3	84.5
	Time Performance (FPS)	12	0.18	0.81	15.75

Table 6 Vehicle counting results comparison on GRAM dataset and ATON Testbed

Compared Algorithm		Yang et al. [37]	Santos et al. [30]	Li et al. [19]	BS-based Method [14]	FR-CNN-based Proposed Method	YOLOv2-based Proposed Method
GRAM Dataset	M-30	Miss detection	6	1	0	1	0
		Precision	92.20	98.7	100	98.7	100
		Time	12	5.67	0.225	0.81	22.5
		Performance (FPS)					
	M-30-HD	Miss detection	5	0	0	0	0
		Precision	88.10	100	100	100	100
		Time	12	5.13	0.18	2.12	19.8
ATON Testbed		Performance (FPS)					
	Highway II	Miss detection	3	1	1	1	1
		Precision	92.31	97.92	97.9	97.9	97.9
		Time	12	6.03	0.18	1.17	16.2
		Performance (FPS)					

Table 7 Vehicle counting results comparison on UA-DETRAC sequences

Compared Algorithm		Li et al. [19]	BS-based Method [14]	FR-CNN-based Proposed Method	YOLOv2-based Proposed Method
UA-DETRAC Dataset	MVI-39401	Miss detection	4	3	3
		Precision	96.36	97.27	97.27
		Time	9	1.4	20.2
	MVI-40852	Performance (FPS)			
		Miss detection	2	1	1
		Precision	96.49	98.24	98.24
		Time	9	0.93	17.4
	MVI-40772	Performance (FPS)			
		Miss detection	2	1	1
		Precision	95.74	97.9	97.9
		Time	10	1.4	19.2
	MVI-40775	Performance (FPS)			
		Miss detection	2	1	1
		Precision	94.59	97.3	97.3
		Time	10	1.29	18
	MVI-40793	Performance (FPS)			
		Miss detection	3	2	2
		Precision	96.15	97.4	97.4
		Time	10	1.38	20
		Performance (FPS)			

Table 8 Vehicle counting results comparison on CDnet2014 sequences

Compared Algorithm	Yang et al. [37]	Santos et al. [30]	Li et al. [19]	BS-based Method [14]	FR-CNN-based Proposed Method	YOLOv2-based Proposed Method
Highway	Precision	100	96.3	100	100	100
	Time (FPS)	5.91	9	0.18	1.31	18.9
	Performance					
Intermittenpan	Precision	N/A	N/A	93.3	100	100
	Time (FPS)	N/A	N/A	0.27	0.49	13.95
	Performance					
Streetcorneratnight	Precision	N/A	N/A	95.2	95.2	95.2
	Time (FPS)	N/A	N/A	0.225	0.63	14.85
	Performance					
TramStation	Precision	N/A	N/A	91.6	91.6	91.6
	Time (FPS)	N/A	N/A	0.18	0.72	15.3
	Performance					

As clear from Tables 6 and 7, the method using YOLOv2 and tracking information achieved the fastest processing speed with comparable counting precision compared to BS-CNN and Faster R-CNN in the detection phase, for GRAM Dataset, ATON Testbed, and UA-DETRAC sequences. It is asserted that the YOLOv2 based detection proposed method increases the time performance by 96.4%, 89.3%, 92.8%, 93%, 94.6%, 92.7%, 92.8%, and 93.1% for M-30, M-30-HD, Highway II, MVI-39401, MVI-40852, MVI-40772, MVI-40775, and MVI-40793, respectively, with the best counting precision accuracy. Authors in [30], proposed a new method to detect, track, and count vehicles by utilizing the detector YOLOv3 and the tracker Deep SORT (Simple Online Realtime Tracking). However, their algorithm needs more time (5.61 FPS) compared to our strategy (19.2 FPS) as shown in Table 6, because YOLOv2 with optical flow information is faster than YOLOv3 and Deep SORT tracker [23]. Table 8 presents the CD net 2014 counting comparison for the proposed strategies in three cases, the first one using background subtraction based CNN in the detection part, the second case using faster R-CNN, and in the third case, using YOLOv2 based detection. The method using YOLOv2 and tracking information achieved the faster processing speed with the best counting precision with background subtraction based CNN and faster R-CNN in the detection phase in TramStation, and Intermittenpan sequences. It is asserted that the YOLOv2 based detection increases the time performance by 99%, 98.1%, 95.8, and 95.3 compared to BS-CNN based detection for Highway, Intermittenpan, Streetcorneratnight, and TramStation respectively. However, using the background subtraction based CNN detection case, the algorithm has the best counting precision accuracy in Intermittenpan, and TramStation sequences.

To investigate the influence of tracking information in our proposed algorithm, we evaluate the proposed algorithm in two cases; the first case, using only the YOLOv2 detection information without tracking information, and in the second one using both the detection and tracking information. We test the CDnet 2014 sequences using experiment III for training YOLOv2. Table 9 shows the comparison result for the two cases.

It is asserted that from Table 9, the proposed method based on collaboration between YOLOv2 and tracking information, increases the time performance by 90%, 87.7%, 88.8%, and 89.7% , 97.6%, 97.5%, 97.8% for Highway, Intermittenpan, Streetcorneratnight, Tram-Station, M-30, M-30-HD, and Highway II, respectively. Since the detection process takes a large time compared to the tracking process, if the detection decision was taken based only on every frame's detection without tracking information, the algorithm takes more time than the detection only once every ten frames. In the latter method, the detection in the remaining frames depends on the tracking information.

Moreover, some missed vehicles have occurred when depending only on the detection information using YOLOv2, such missed vehicles can be detected through the frames once they detected one time through the frames using the tracking information. As shown in Table 9, the YOLOv2 with tracking information improves the Recall accuracy by 7.5%, 4%, 6%, 3%, 5%, 6%, and 7% for Highway, Intermittenpan, Streetcorneratnight, TramStation, M-30, M-30-HD, and Highway II, respectively.

5 Conclusion

In this work, efficient vehicle detection and counting scheme based on convolution neural networks (CNNs) and KLT tracker was introduced. YOLOv2 classifier detector was used

Table 9 Vehicle detection results comparison between YOLOv2 based only and YOLOv2 with tracking information

Method		YOLOv2	YOLOv2-based
Sequence videos		Based Decision	Proposed Method
Highway	Precision %	100	100
	Recall %	84.5	92
	Time Performance (FPS)	1.89	18.9
Intermittenpan	Precision %	100	100
	Recall %	86	90
	Time Performance (FPS)	1.71	13.95
Streetcorneratnight	Precision %	100	100
	Recall %	74	80
	Time Performance (FPS)	1.67	14.85
Tramstation	Precision %	100	100
	Recall %	87	90
	Time Performance (FPS)	1.57	15.3
M-30	Precision %	100	100
	Recall %	91	96
	Time Performance (FPS)	0.54	22.5
M-30-HD	Precision %	100	100
	Recall %	88	94
	Time Performance (FPS)	0.49	19.8
Highway II	Precision %	100	100
	Recall %	89	96
	Time Performance (FPS)	0.36	16.2
Average accuracy	Precision %	100	100
	Recall %	85.6	91.1
	Time Performance (FPS)	1.17	17.35

for vehicle detection. The detection process was performed in the first frame every frame-set (N-frames). Then, the KLT tracker had been adopted with K-means clustering to refine and cluster the detected vehicles through the N-frames. Finally, merging clusters was introduced to classify the vehicles detected with its correspondence vehicles cluster. Different experiments are conducted to investigate pixel CNN based classification and regression CNN based classification, including faster R-CNN and YOLOv2 for vehicle detection and counting. In addition, we investigate the algorithm when the detection is based only on the YOLOv2 algorithm compared to the detection decision based on YOLOv2 with the tracking information. The proposed method is faster than the background subtraction and faster R-CNN based detection. To achieve good detection and counting results using YOLOv2, the training images must include different samples of the vehicle scenes as noticed in experiment III and experiment II. Moreover, It has been presented that the tracking information improved the average Recall accuracy by 5.5% and improved the time performance by an average of 93.3% for different seven sequences. The proposed methods handled the state

of art disadvantages accompanied by traditional and CNN's approaches, with low computational complexity, better detection and counting accuracy, and working with different and complex traffic scenes.

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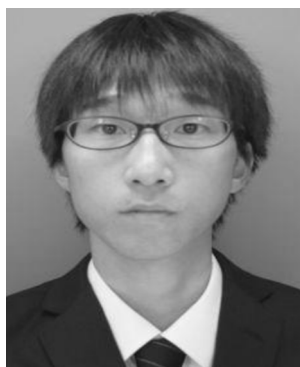
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